

Name: _____

Homework #05: RAM & ROM

Directives

1. Write the **directive** to create a **32-bit** constant named **const1** initialized to **1**.
2. Write the **directive** to create a **16-bit** constant named **const2** initialized to **0x2**.
3. Write the **directive** to create an **8-bit** constant named **const3** initialized to **2_11** (binary 3).
4. Write the **directive** to create a **32-bit** variable named **var4**.
5. Write the **directive** to create a **16-bit** variable named **var5**.
6. Write the **directive** to create an **8-bit** variable named **var6**.

Load/Store Registers from/into RAM/ROM

7. Write the assembly code to **load** register **R0** with the value of the **32-bit** constant named **const1**.

8. Write the assembly code to **load** register **R0** with the value of the **16-bit** constant named **const2**.

9. Write the assembly code to **store** the value in register **R0** to the **32-bit** variable named **var4**.

10. Write the assembly code to **store** the value in register **R0** to the **8-bit** variable named **var6**.

Programming

11. Write the assembly program to accomplish the following C instruction: (Both const1 and var4 are 32 bits and they already defined and initialized.)

```
Var4 = const1;
```

12. Write the assembly program to accomplish the following C code: (varY is 32 bits and already defined and initialized).

```
While (varY<10) {  
  
    varY++;  
  
}
```